Randomized Experiments and Ad Measurement

MKT 566 · Fall 2026

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Before We Start

Housekeeping

- **Homework 4** is out, due **Nov. 24** on Brightspace
- Today: **randomized experiments** and how firms measure advertising
- Thursday: causality when you **cannot** run an experiment (observational data)
- Next Tuesday, Nov. 17: in-class **causality exercise**, bring your laptop
- Thursday, Nov. 19: class time to **work on your group project** with your team
- Final presentations: **Dec. 1 and Dec. 3**

Where we are in the course

- Weeks 3 to 11 were about **prediction**: regression, logit, trees, forests, text. All of them answer "what is likely to happen?"
- The next two weeks are about a different question: **"did my action cause the change?"**
  - Did the coupon raise sales?
  - Did the ad campaign bring in customers who would not have come anyway?
  - Did the new badge on the product page sell more units?
- A model that predicts well does **not** answer these. Today we see why, and what does

Stats Refresher

Averages from samples bounce around

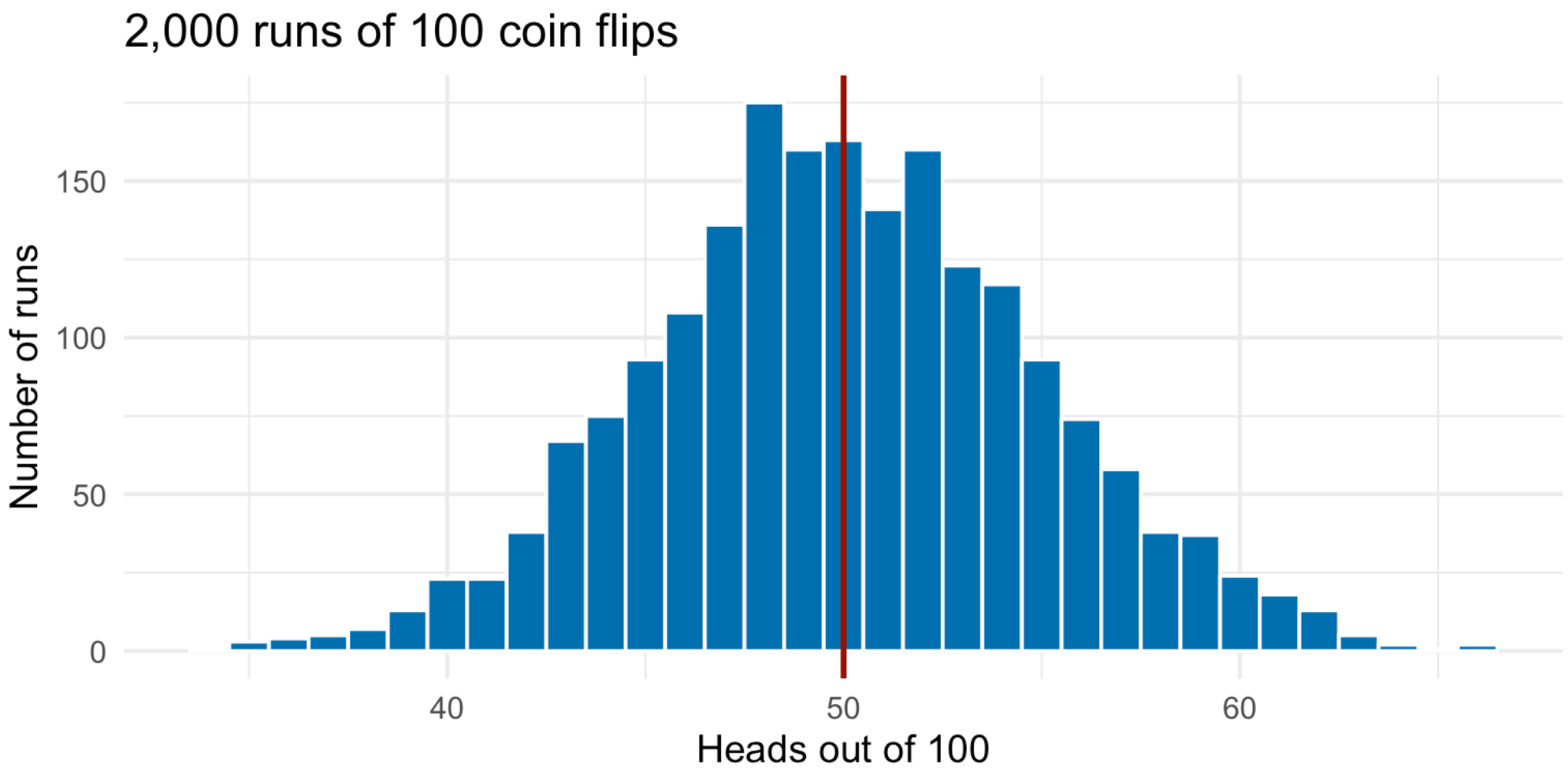
Today we compare groups: coupon vs. no coupon, subject line A vs. B. Groups always differ a little, even when nothing is going on. Start with a coin.

Flip a fair coin 100 times. The truth is 50% heads. Five runs on the computer:

Run	Heads out of 100	Share of heads
1	54	54%
2	53	53%
3	49	49%
4	51	51%
5	53	53%

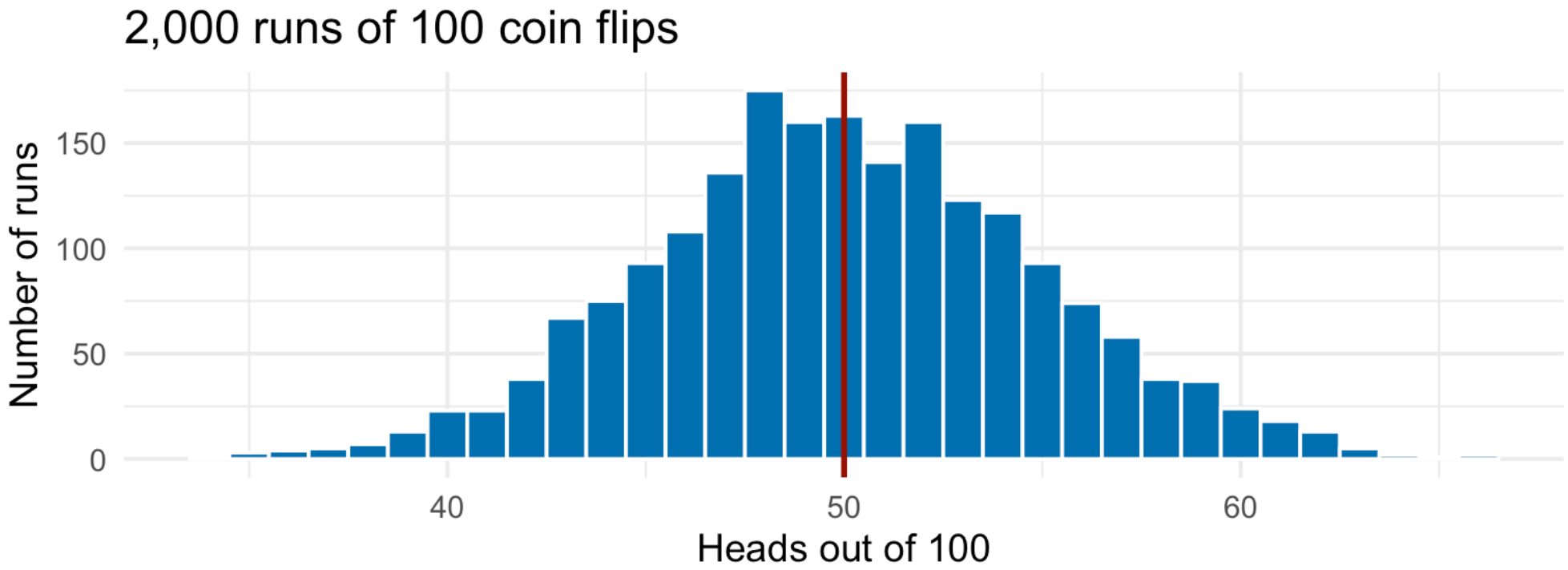
- No run gives exactly 50. The share moves around by **luck of the draw**, although the coin never changed
- Same with customers: the average spend of 1,000 random customers is not the average of all customers, and another 1,000 would give a different number
- This bounce is called **sampling noise**. The next slides measure it

How much does it bounce?



We repeated the 100 flips 2,000 times and counted heads each time. The red line is the truth, 50. What do you see?

How much does it bounce? The standard error



- The runs pile up around 50 and spread out by about **±5 heads** either way. That typical distance from the truth is the **standard error**: here 5 heads, or 5 percentage points
- Nobody runs 2,000 experiments to get it. A formula gives the same number: $\sqrt{(p \times (1 - p) / n)} = \sqrt{(0.5 \times 0.5 / 100)} = \sqrt{0.0025} = 0.05$ = 5 points
- Same formula with 10,000 flips: 0.5 points. **More data, less noise**: 100 times the data cuts the noise by 10 (the square root)

The 95% rule, and the confidence interval

- In the histogram, about 95% of the runs (94% in this simulation) land within **2 standard errors** of the truth: 50 ± 2 × 5 = 40 to 60 heads (the exact multiplier is 1.96, which is why 1.96 shows up everywhere)
- Now flip it around. In real life you have **one** run and do not know the truth. But “my result ± 2 standard errors” is a range that contains the truth 95 times out of 100. That range is the **95% confidence interval**
- Example: run 1 gave 54 heads. 54 ± 2 × 5 = 44 to 64. The range contains 50, so nothing says the coin is unfair

Is a difference real? The p-value

- Two groups, two averages, a gap. Start from the boring explanation: **there is no real difference**, the gap is sampling noise
- The **p-value** is how often sampling noise alone would produce a gap at least this big
- Small p-value, below **0.05** by convention (fooled by luck 1 time in 20): noise is a bad explanation, the gap is **statistically significant**
- Shortcut: gap ÷ standard error = how many standard errors away. Beyond about 2, p is below 0.05
- You saw all of this in week 4: coefficient, standard error in parentheses, stars for $p < 0.05$ ($*$), 0.01 ($**$), 0.001 ($***$). Thursday's tables use the same format

> Everything that follows today is this logic applied to one comparison: treatment group vs. control group.

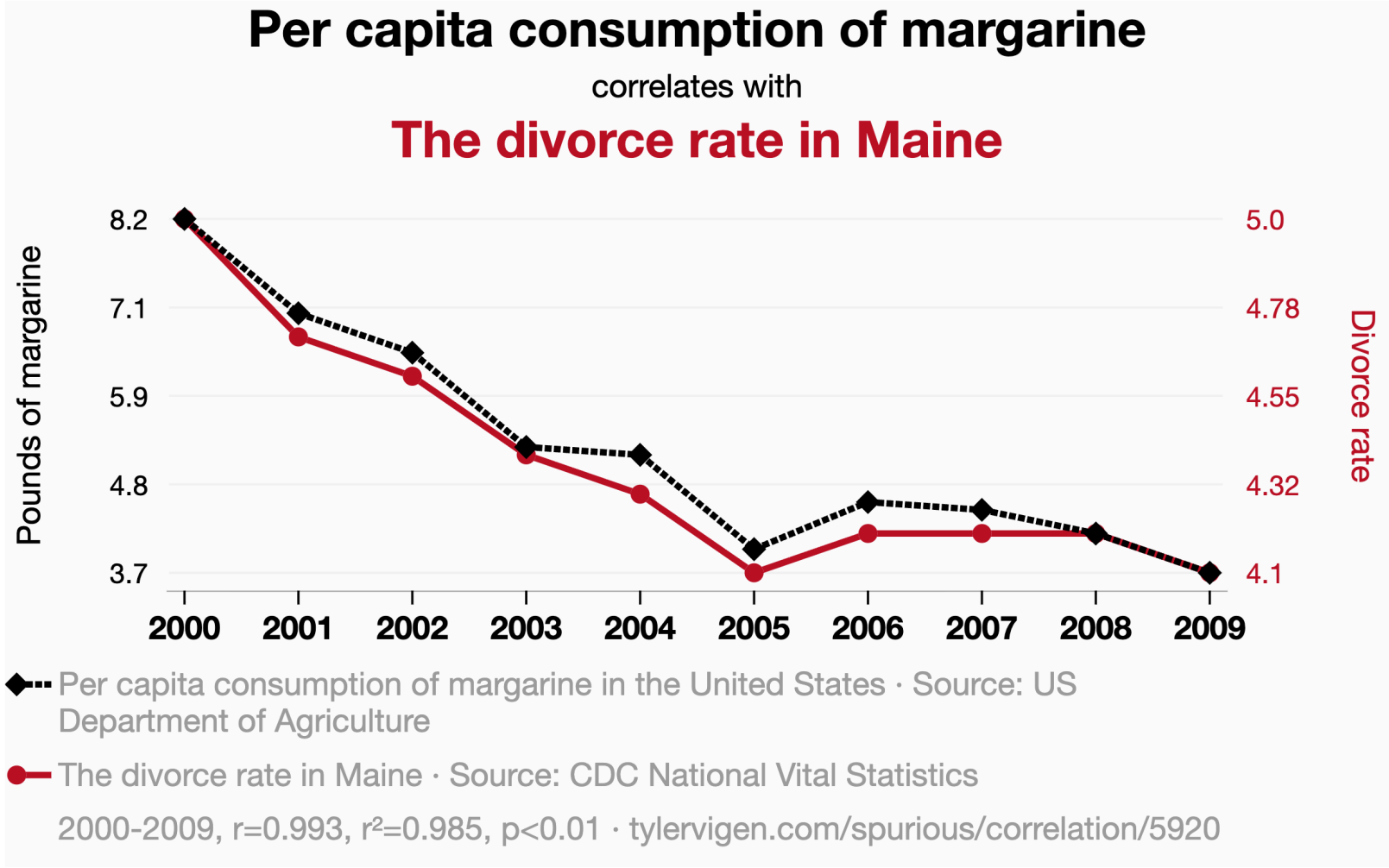
Correlation Is Not Causation

The question every marketer asks

"We ran the campaign and sales went up. Did the campaign do it?"

- The data can show you a **correlation**: when spend was high, sales were high
- A correlation tells you one thing: **the two variables move together**. Anything beyond that requires assumptions
- Sales may have gone up because of the campaign, or because of the season, or because the campaign was launched **when sales were already rising**
- All three stories produce the same correlation. The correlation cannot tell them apart

Two lines that move together



What do you see? Does eating less margarine save marriages in Maine?

Two lines that move together, explained

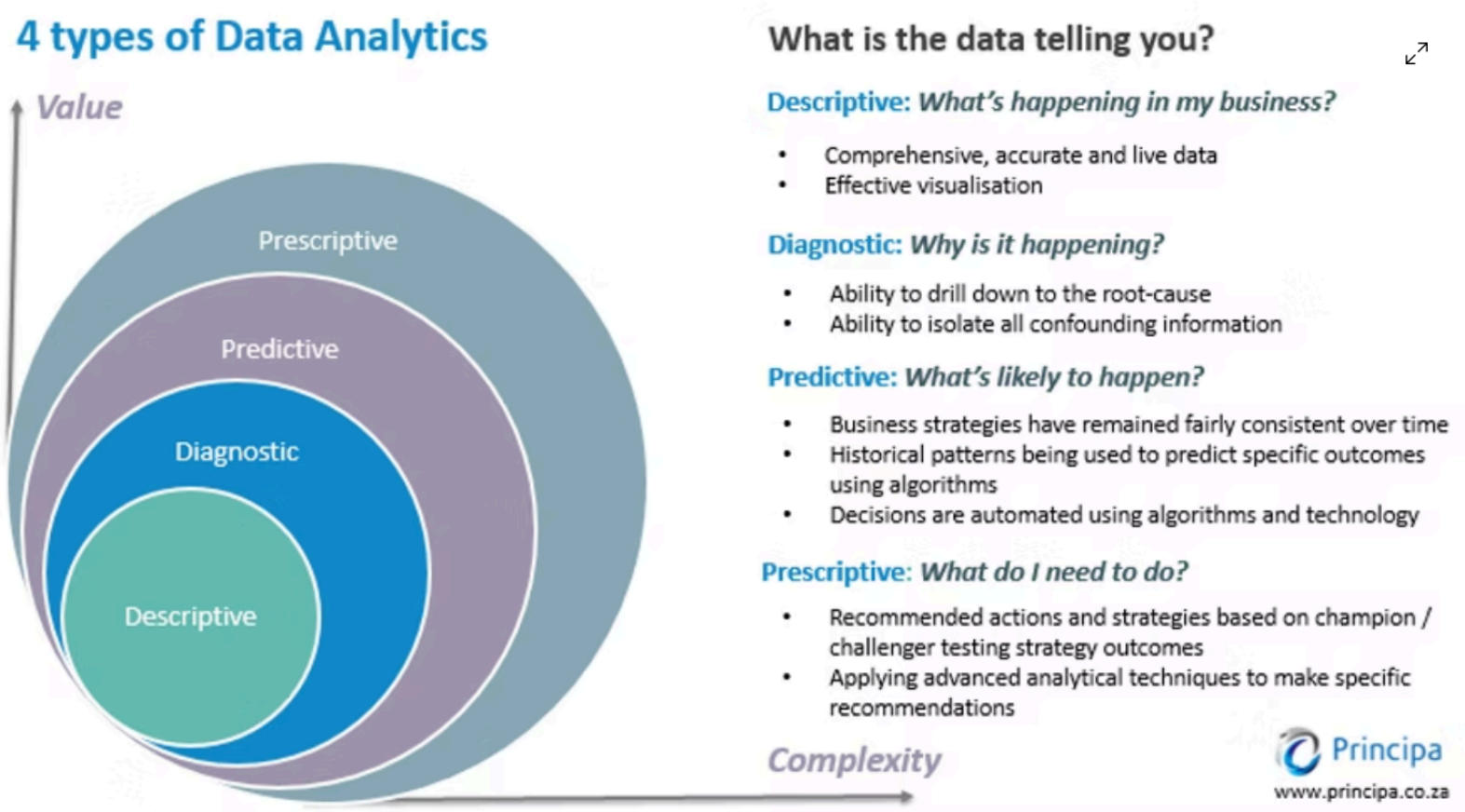
- The correlation is 0.99, as strong as it gets. And there is **no causal link** at all
- With thousands of series to choose from, some pairs move together **by chance**. Pick enough pairs and you will always find one
- Three ways two variables can move together without one causing the other:
  - **Chance**, as here
  - **A third factor** drives both (a hidden “cause of both”)
  - **Reverse causality**: B causes A, not A causes B

Three classic traps

What the data show	What is really going on
People carrying **umbrellas** on days it rains	Umbrellas do not cause rain. People check the forecast and act **ahead of** the rain: forward-looking behavior
Kids who get **tutoring** have lower grades	Tutoring does not lower grades. The kids who get tutoring are the ones who were **already struggling**: selection
Ice cream sales and **drownings** rise together	Neither causes the other. **Summer** drives both: a hidden third factor

> In every marketing version of these traps, the "treatment" (the ad, the coupon, the feature) was **not handed out at random**. Somebody chose who gets it, and that choice is correlated with the outcome.

Why causality matters



- **Descriptive** ("what is happening?") and **predictive** ("what is likely to happen?") run on correlations. If umbrellas predict rain, a forecaster can use umbrellas
- **Diagnostic** ("why?") and **prescriptive** ("what should we do?") need causality. Handing out umbrellas will not make it rain

The Counterfactual

The one thing you can never see

Maria got the coupon and spent **$60**. Did the coupon work?

- To answer, you need to know what Maria would have spent **without** the coupon. That world never happened. It is called the **counterfactual**
- The effect of the coupon on Maria is $60 minus the counterfactual. One of the two numbers is **missing, for every single customer, always**
- This is a **missing-data problem** that no amount of extra data on Maria can fix: you only ever get one of her two worlds

> Causal inference is the art of filling in the missing world **credibly**.

Every customer is missing half the table

Customer	Got the coupon?	Spend with coupon	Spend without coupon	Effect
Maria	yes	$60	?	?
James	yes	$45	?	?
Priya	no	?	$38	?
Tom	no	?	$52	?

- The effect for one person needs both columns. Nobody has both
- The way out is to stop asking about individuals and ask about **groups**: compare the **average** spend of the coupon group with the average of the no-coupon group
- That only works if the two groups are **the same kind of people**, so that the no-coupon group's average is a fair stand-in for the coupon group's missing column

Two ways to hand out the coupon

2,000 customers (simulated). We assign the coupon two ways and compare the groups **before** the coupon does anything:

How the coupon was assigned	Group	customers	avg. past spend	avg. age
Coin flip (random)	No coupon	992	$94	38.6
Coin flip (random)	Coupon	1008	$96	37.9
Targeted at big spenders	No coupon	1367	$75	38.1
Targeted at big spenders	Coupon	633	$137	38.5

What do you see? In which case would “coupon group average minus no-coupon group average” be a fair measure of the coupon?

Two ways to hand out the coupon, explained

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- **Coin flip**: the two groups match on past spend (a gap of $2) and on age. They also match on everything we did **not** measure, because a coin does not look at the customer
- **Targeted**: the coupon group was already spending $61 more **before** any coupon. Any sales difference afterwards mixes two things: the coupon, and the fact that these were bigger spenders to begin with

> **Randomization** is what makes the no-coupon group a credible stand-in for the missing world.

The logic of an experiment

- Split the units (customers, users, stores) **at random** into a treatment group and a control group
- Treat one group, not the other. Everything else is held constant **on average**, because chance decided who went where
- Compare the average outcome in the two groups. The difference is the **causal effect** of the treatment, plus some noise from the coin flips

> This is the whole idea. Everything else today is about doing it well, and about what happens when firms skip it.

Anatomy of a randomized experiment

Element	What it is	Running example
Population	the units you randomize	20,000 email subscribers
Treatment	the one thing you change	a new subject line
Randomization	how units are assigned	50/50 coin flip per subscriber
Outcome	what you measure	opened the email (yes/no)

- In practice, a **randomized controlled trial** (RCT) in marketing is just this table with the cells filled in
- The hardest cell is often the second one: the treatment should change **one thing**, or you will not know which change did the work

Randomization in practice

- The **unit** of randomization matters: user, session, store, city, campaign. Pick the unit at which the treatment is delivered and at which outcomes are measured
- Always **check balance**: compare pre-treatment variables (past spend, age, device, region) across groups, exactly like the coin-flip vs. targeted table in the coupon example (no coupon $94 vs. coupon $96 in past spend, same age). Randomization produces balanced groups on average; a specific draw can still come out unbalanced
- If the groups are **not** balanced, suspect the randomization before you trust the result (the “random” split in many companies is a rule like “odd user IDs”, which is not always random)

A/B Testing

A/B test: the simplest experiment

- Two versions, **A** and **B**, that differ in **one element**. Users are randomly shown one or the other. Measure the outcome, pick the winner
- It is an RCT with a product name. Used for web design, pricing, ad creatives, subject lines, recommendation algorithms
- Large platforms run thousands at a time; a newsletter can run one with a built-in button

> The next five slides are one A/B test, start to finish, with every number worked out.

One A/B test, worked out

Two subject lines. 10,000 subscribers get A, 10,000 get B.

	Subject line A	Subject line B
Emails sent	10,000	10,000
Opened	2,130	2,310
Open rate	21.3%	23.1%

- B opened 1.8 pp more often (23.1% minus 21.3%)
- Before declaring B the winner: **could a gap this size happen by chance**, with the two subject lines equally good?

Is B really better, or just lucky?

- Flip two fair coins 10,000 times each. They will **not** land heads the same number of times. The difference is luck
- Same for emails: even if A and B were **exactly the same**, their open counts would differ a little, by luck
- So the question is: **how big a gap can luck alone make?** We find out by running fake tests on the computer

One fake test:

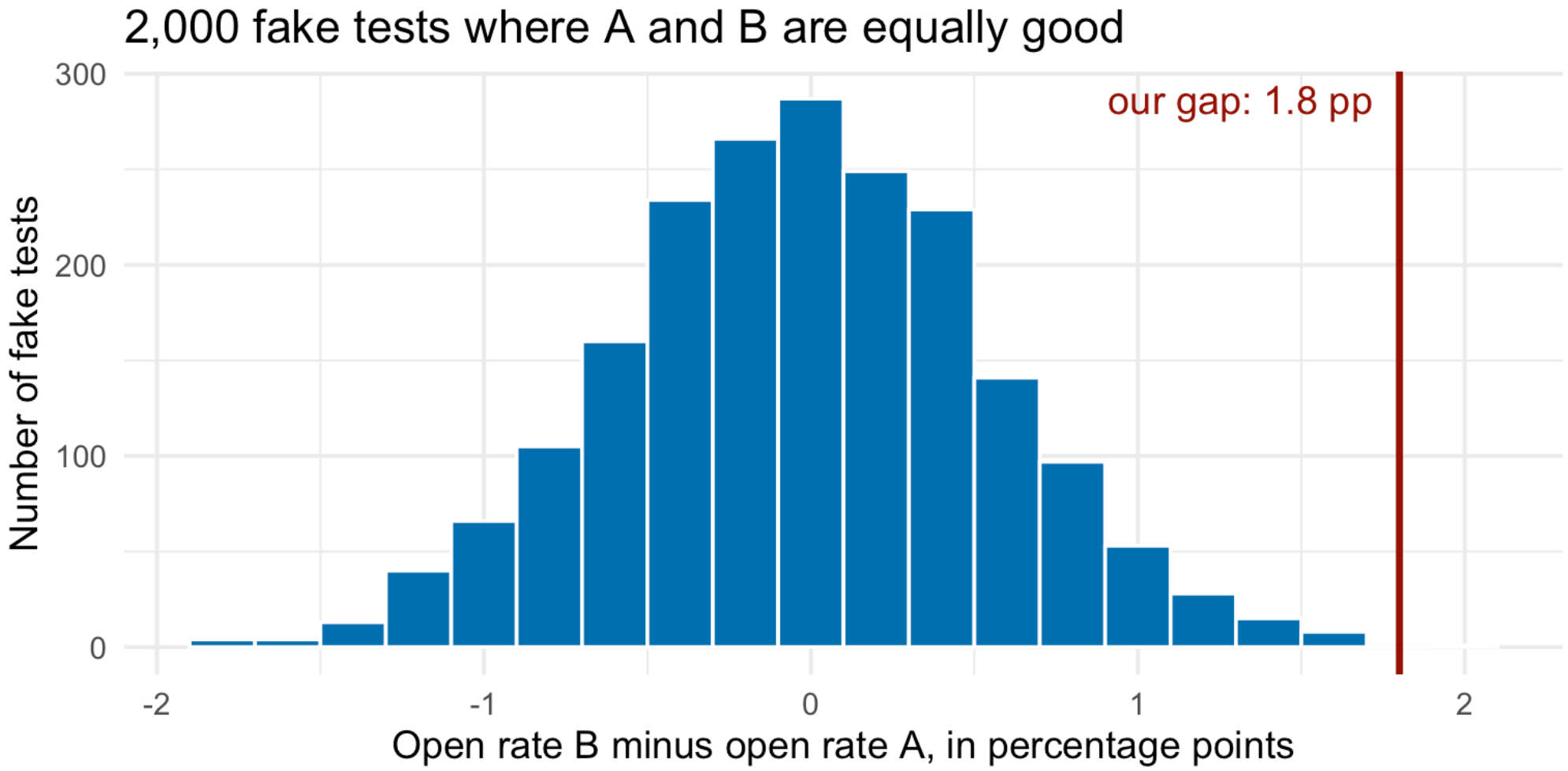
1. Every email gets the **same** 21.3% chance of being opened, whatever the subject line (that is A's open rate)
2. The computer "sends" 10,000 emails as A and 10,000 as B, and rolls the dice for each one: opened or not
3. Count the opens in A and in B. The gap between them is **pure luck**, because the two groups were identical

We do this 2,000 times. The first three fake tests:

Fake test	A opened	B opened	Gap, B minus A (pct. points)
1	2,104	2,076	-0.28
2	2,092	2,176	+0.84
3	2,108	2,095	-0.13

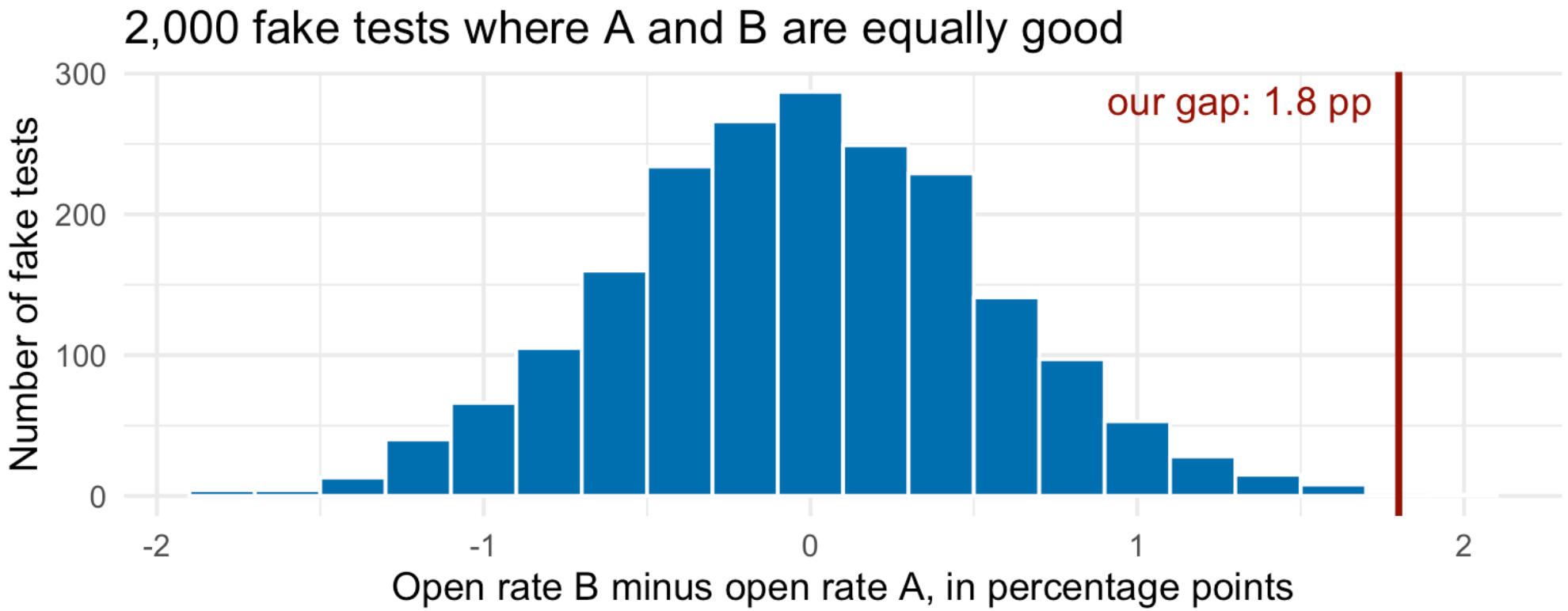
> Next: how often does luck alone make a gap as big as our 1.8 percentage points (23.1% minus 21.3%)?

What chance alone produces



Each bar counts how many of the 2,000 fake tests produced that gap. The red line is the gap we actually observed. What do you see?

What chance alone produces, explained



- Luck alone does make gaps, but small ones: 95 out of 100 fake tests have a gap between -1.1 and 1.1 percentage points
- A gap as big as ours (1.8 percentage points) showed up in only **3 of 2,000** fake tests. That fraction (3/2,000 = 0.15%) is what a **p-value** is: how often luck alone produces what we saw
- Our gap is far outside what luck produces. B is better

The same thing, with a formula

The simulation is what a **hypothesis test** does with a formula instead of 2,000 fake tests:

Step	Question	Arithmetic	Result
1. Null hypothesis	what if A and B are equally good?	gap = 0	
2. Difference	how big is the observed gap?	23.1% − 21.3%	1.8 pp
3. Standard error	how much does chance move the gap?	$\sqrt{(p_A(1-p_A)/n_A + p_B(1-p_B)/n_B)}$	0.59 pp
4. z	how many standard errors is the gap?	1.8 / 0.59	3.06
5. p-value	how often would chance produce this?	from z	0.002
6. 95% confidence interval	which gaps are plausible?	1.8 ± 1.96 × 0.59	0.6 to 3.0 pp

- The **95% confidence interval** is gap ± 1.96 standard errors. Read it as: the true gap is plausibly anywhere from 0.6 to 3.0 percentage points. It says how big the effect is, which the p-value does not
- Where 1.96 comes from: ± 1.96 standard errors covers 95% of what luck alone produces (1.96 × 0.59 = 1.2, the -1.1 to 1.1 range of the fake tests)
- **Small samples are noisy.** Same open rates with 1,000 emails per arm: standard error 1.86 pp, z = 0.97, p = 0.33. The same gap, and no conclusion

Common mistakes in A/B testing

- **Peeking**: checking the test every day and stopping the moment B looks ahead. Remember the fake tests: even when A and B are identical, B is ahead on many days just by luck. Stop on one of those days and you crown a winner that is not one. Decide the number of emails in advance and read the result once, at the end
- **Too short, too small**: with 1,000 emails per arm, our same 1.8-point gap gave p = 0.33 (previous slide). The test could not tell luck from a real effect. “Not significant” then means “we could not tell”, not “no effect”
- **Many tests at once**: the rule “call it a winner when $p < 0.05$” means we accept being fooled by luck 1 time in 20. So if you test 20 changes that do nothing, about 1 of them will look like a winner by luck alone. Expect it, and re-test before you ship it
- **Spillovers**: the control group must not be touched by the treatment. If treated users talk to control users (a referral bonus), or compete with them for the same drivers or inventory (a discount on rides), the control group changes too, and the comparison is off
- **Significance is not value**: p-values get small when the sample is huge. With 10 million users, a gain of 0.01 points is “significant”. Significant means “not luck”, not “worth it”. Whether to ship it is a business question: what is 0.01 points worth in dollars?

Ad Measurement

What ad measurement is, and why it is hard

- **Ad measurement**: figuring out what advertising did to the outcomes you care about (awareness, clicks, conversions, sales)
- It is hard because the effect of an ad depends on its content, where it ran, when, to whom, on market conditions, and on **all the ads that came before**
- It is **expensive**. So a measurement should exist to inform a decision: which channel gets the budget, whether to keep the campaign. A number that will not change a decision is money wasted

The number everyone reports: ROAS

A campaign spends **$50,000**. The platform's dashboard **attributes** 1,800 conversions to it, each worth $100.

- Attributed revenue: 1,800 × $100 = **$180,000**
- **ROAS** (return on ad spend) = (attributed revenue − spend) / spend = ($180,000 − $50,000) / $50,000 = **2.6**

A return of 2.6 dollars per dollar. Great campaign?

> The question the dashboard does not answer: **how many of those 1,800 people would have bought anyway**, with no ad? "Attributed" means the person saw or clicked the ad before buying. That is a correlation.

Attribution vs. incrementality

	Attribution	Incrementality
The question	which ad or channel gets **credit** for a conversion?	how much of the behavior **would not have happened** without the ad?
The data	observational: click paths, impressions	experimental: holdout tests, ghost ads, geo tests
Typical output	“search 40%, display 30%, social 30%”	lift in %, incremental conversions, **iROAS**
Limitation	correlational: cannot tell whether the ad **caused** anything	needs randomization, which is harder to set up

Same word, two meanings: **ROAS** uses attributed revenue; **incremental ROAS (iROAS)** uses only the revenue the ads caused. They can be very different numbers.

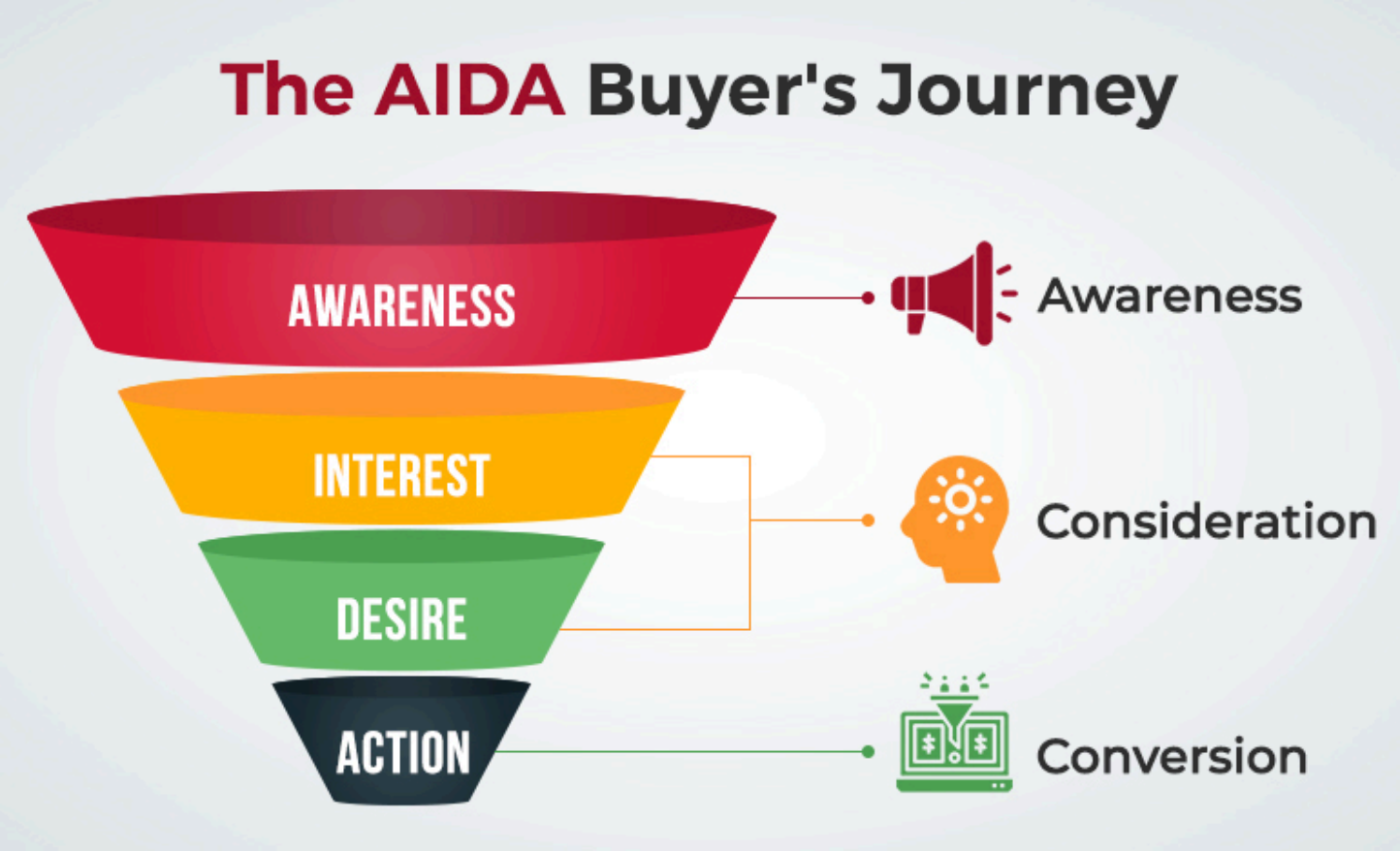
Attribution rules, on one customer

One customer sees a search ad on day 1, a social ad on day 3, gets an email on day 5, and buys for $100. Who gets the credit?

Rule	Search ad (day 1)	Social ad (day 3)	Email (day 5)
Last-touch	$0	$0	$100
First-touch	$100	$0	$0
Linear	$33	$33	$33
Time-decay	$20	$30	$50

- Every rule gives a **different answer** for the same purchase, and all of them add up to $100
- Not one of them asks whether the customer **would have bought anyway**. If she would have, the right answer is $0 for everyone
- Easy to compute, so widely used; but they allocate credit, they do not measure cause

Ads work along a funnel



- Short-run responses (clicks, purchases this week) are **much easier** to measure than long-run ones (awareness, brand preference)
- Most incrementality tests measure the bottom of the funnel. Keep in mind what they leave out

Measuring incrementality: the holdout test

Same campaign as the ROAS slide: **$50,000** spent, **1,800** attributed conversions, $100 each. The audience is 100,000 people. This time we randomly **withhold** the ads from 10,000 of them. That slice is the control group.

	Treatment (ads turned on)	Holdout, the control group (ads turned off)
Users	90,000	10,000
Conversion rate	3.2%	2.8%

- Lift = 3.2% − 2.8% = **0.4 pp**
- Incremental conversions = 0.4 pp × 90,000 users = **360**, not 1,800
- Incremental revenue = 360 × $100 = $36,000
- **iROAS** = ($36,000 − $50,000) / $50,000 = **-0.28**

> Same campaign. The dashboard said 2.6; the experiment says the campaign **lost money**. The 1,440 other buyers were going to buy anyway.

The catch with holdouts

Same campaign as the holdout slide. “Ads turned on” does not mean “sees the ads”: most people in the treatment group **never see an ad**, because an ad only appears if the person visits a site where your bid wins the auction. Say that happens to 20% of them (18,000 of the 90,000), and they are not a random 20%: they are the heavy internet users.

Kind of person	Share of the audience	Converts without ads	Converts if shown the ad
Heavy internet users (the ad reaches them)	20%	4.0%	6.0%
Light users (the ad never reaches them)	80%	2.5%	2.5% (never shown)

What is the true causal effect of seeing the ad? Which two numbers in the table give it?

The catch with holdouts, explained

Same campaign as the holdout slide. “Ads turned on” does not mean “sees the ads”: most people in the treatment group **never see an ad**, because an ad only appears if the person visits a site where your bid wins the auction. Say that happens to 20% of them (18,000 of the 90,000), and they are not a random 20%: they are the heavy internet users.

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The true effect of seeing the ad is **2.0 pp** (6.0% − 4.0%): the same people, with and without the ad. We know it because we made up the numbers. Can we recover it from the data? Three attempts.

Attempt 1: whole treatment group vs. whole control group

The obvious choice, and what we did on the holdout slide: compare everyone in the treatment group (ads turned on, whether they saw one or not) with everyone in the holdout (ads turned off). Same two kinds of people as before:

Kind of person	Share of the audience	Converts without ads	Converts if shown the ad
Heavy internet users (the ad reaches them)	20%	4.0%	6.0%
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- Treatment group: heavy users saw the ad, light users did not. Average rate = 20% × 6.0% + 80% × 2.5% = 3.2%
- Holdout: nobody saw an ad. Average rate = 20% × 4.0% + 80% × 2.5% = 2.8%
- Gap: 3.2% − 2.8% = **0.4 pp**, the number on the holdout slide. Fair, because the groups were randomized, and right in total: 0.4 pp × 90,000 = 360 incremental conversions
- But the ad only did something to the 18,000 people who saw it. Its 2.0 pp effect is **diluted** across all 90,000: 2.0 × 20% = 0.4. A tiny gap, hard to tell from sampling noise unless the sample is enormous. So analysts look for something sharper

Attempt 2: exposed vs. not exposed

The tempting fix for the dilution: inside the treatment group (ads turned on), compare only the people who actually saw an ad with the people who never did:

Kind of person	Share of the audience	Converts without ads	Converts if shown the ad
Heavy internet users (the ad reaches them)	20%	4.0%	6.0%
Light users (the ad never reaches them)	80%	2.5%	2.5% (never shown)

- Saw the ad: 6.0% convert. Did not see it: 2.5%. Gap: **3.5 pp**
- Wrong, and it is the most common number in ad reports. The two groups are **different kinds of people**: heavy users convert at 4.0% with no ad at all, light users at 2.5%. The gap is mostly who they are, not the ad
- 3.5 pp is 1.7 times the true 2.0 pp. This is **selection**, the same problem as the targeted coupon

Attempt 3: exposed vs. the control users who would have been exposed

Kind of person	Share of the audience	Converts without ads	Converts if shown the ad
Heavy internet users (the ad reaches them)	20%	4.0%	6.0%
Light users (the ad never reaches them)	80%	2.5%	2.5% (never shown)

- Compare the heavy users who saw the ad (6.0%) with the heavy users in the holdout, who **would have seen it** had ads been turned on for them (4.0%)
- Gap: **2.0 pp**. Right: same kind of people, and the only difference between them is the ad
- Same 360 conversions as Attempt 1 (2.0 pp × 18,000 exposed = 360), but measured on the 18,000 people the ad could affect instead of spread over 90,000: a gap five times bigger, much easier to tell from noise
- The problem: nobody in the holdout saw an ad, so **how do you know who would have been exposed?**

> That question is what the “ghost ads” design answers.

The “ghost ads” design

The answer to “who would have been exposed”: run the auction for the control group too, just do not show the ad.

- Users are randomized into treatment and control **before** any ad is served
- The ad auction runs for everyone. When your ad wins for a treatment user, it is shown. When it wins for a control user, the platform **logs it** and shows something else: a “ghost” impression
- Now the comparison is exposed treatment users vs. ghost-exposed control users: **the same kind of people**, selected by the same auction. In our example: 6.0% vs. 4.0% = 2.0 pp on 18,000 people, instead of 0.4 pp on 90,000
- Used by Amazon, Meta, and Google for their lift studies. Needs the platform’s cooperation, because only the platform sees the auction

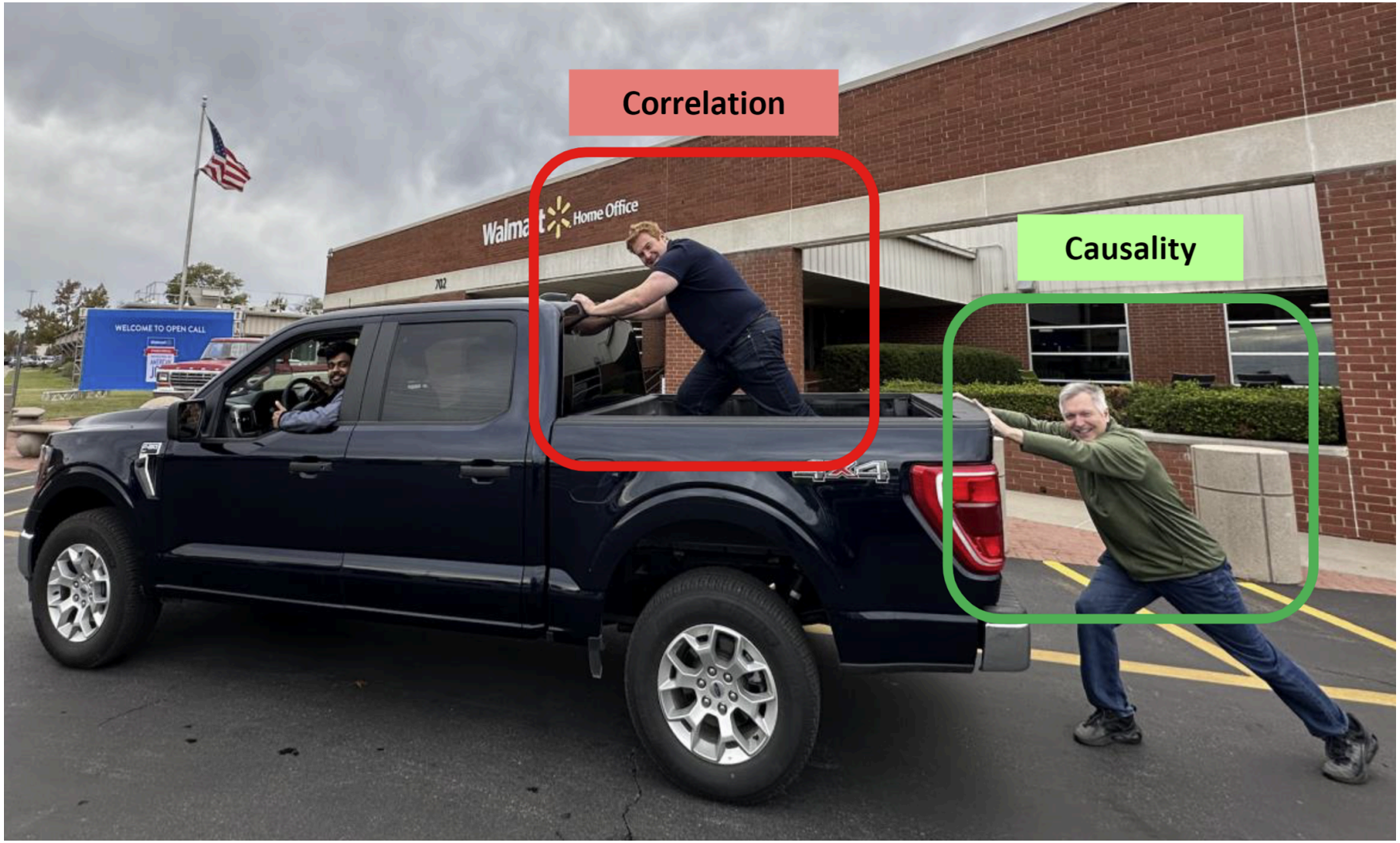
Geo-split tests

When you cannot randomize people (TV, radio, billboards, campaigns with no user-level tracking), randomize **places**:

- Pick comparable cities or regions. Run the campaign in a random half, hold it back in the other half, compare sales
- Works for brand and offline campaigns, where user-level holdouts are impossible
- The cost: you only have as many observations as you have markets (20 cities, not 200,000 users), so the test needs **large effects** or **many comparable markets** to see anything

Why Firms Still Use Correlations

Who is moving the truck?



Both men are moving forward with the truck. Which one is making it move?

Who is moving the truck, explained

- The man in the truck bed moves **with** the truck: perfect correlation, zero causal contribution. The man behind is doing the work
- A correlational ad estimate is the man in the bed: ad spend and sales move together, and it looks like pushing
- Some firms accept this, and sometimes they get away with it: a correlation **might** point in the right direction, and a biased estimate **might** still lead to the right decision
- But direction is only half the picture. Budget decisions need the **size** of the effect, and that is where correlations go badly wrong (the 2.6 vs. -0.28 from before)

The incentives behind the numbers

Why correlational ad numbers survive, even at firms that know better:

- Measuring causal effects is **hard**: many firms lack the expertise, the discipline, or the execution skill; small tests come back inconclusive; tests get designed with no decision attached, so the result changes nothing
- **Platforms** provide the correlational estimates themselves, and they are “black box”: no outside auditor checks them
- Correlational estimates are **larger** than experimental ones (they count the people who would have bought anyway). Which number does a marketer prefer to show the boss?
- **Agencies** have historically been paid on spend, not on incremental sales
- “Nobody ever got fired for buying [big platform]”

> Attribution is a system for **maximizing the credit given to ads**. Incrementality is a system for measuring what ads did. Know which one a number comes from.

Marketing mix models (MMM)

A different, older approach: no experiment, no user-level data, just **weekly totals** and a regression.

$$\text{Sales}_t = \beta_0 + \beta_1 \text{TV}_t + \beta_2 \text{Search}_t + \beta_3 \text{Display}_t + \cdots + \text{controls}_t + \varepsilon_t$$

- Each β is the average **correlational** effect of a dollar in that channel, with seasonality, price, promotions, holidays as controls
- Open-source toolkits from the platforms: Google **Meridian**, Meta **Robyn**
- One ingredient worth knowing: **adstock**, the carryover of advertising into later weeks

Week	Spend	Adstock (50% carries over)
1	$100	100
2	$0	50
3	$0	25
4	$0	12.5
5	$0	6.25

MMM: what it is for, and what it cannot do

Used for

- Estimating a return per channel and supporting **budget allocation** across channels
- Settings where user-level data do not exist (TV, radio, print) and experiments are impractical
- Capturing slow, long-term brand effects that a two-week holdout cannot see

Limitations

- It is a regression on observational data: **no randomization**, so the β's are correlations with assumptions on top
- Channels move together (TV and search spend both rise before the holidays), so the model struggles to tell them apart
- Slow feedback: usually refit quarterly or yearly

Wrapping Up

Takeaways

- A correlation says two things move together. A **causal effect** is the difference between what happened and what would have happened otherwise: the **counterfactual** you never observe
- **Randomization** fills in the counterfactual credibly: the control group stands in for the missing world, because a coin flip does not look at the customer
- An **A/B test** is an experiment with a product name. Decide the sample size in advance, do not peek, and remember that “significant” is not the same as “worth it”
- In advertising, **attribution** allocates credit; **incrementality** measures cause. ROAS and iROAS can disagree by a lot, and the incentives favor the bigger number
- Correlational methods (attribution, MMM) remain popular because experiments are hard. Know which kind of number you are looking at

> Thursday: when you **cannot** randomize. Regression with controls, matching, and difference-in-differences, with a real Amazon case.